

Wildlife-Tracking Drone Design

NAME _____ DATE _____ PERIOD _____

1. Read the fictional user need

User: _____

Animal and environment: _____

Three mission needs:

1. _____

2. _____

3. _____

Two environmental or animal-behavior constraints:

1. _____

2. _____

2. Write requirements

Model: Need: locate an animal at night. Requirement: the system must gather usable low-light or thermal evidence and send a location record to the research team.

Flight requirement:

Sensing or data requirement:

Communication requirement:

Environmental-protection requirement:

3. Draw the top-down or side-view blueprint

Label flight, power, navigation, payload/data, communication, and environmental protection.



4. Explain the six system jobs

SYSTEM JOB	COMPONENT OR FEATURE	WHAT IT DOES FOR THIS MISSION
Flight		
Power		
Navigation / obstacle sensing		
Data collection payload		
Communication / data return		
Environmental protection / wildlife disturbance		

One assumption the team would need to test:

One tradeoff in the design:

5. Changed mission: sea turtles at night

The new conservation team works on a remote beach at night and must avoid disturbing nesting sea turtles.

One component I would keep:

Why it still fits:

One component I would change:

Mission evidence that makes the change necessary:

One occupation that would help test, build, operate, or interpret the system:

That person's job in the mission:

Done when

- ☐ Six labeled system jobs
- ☐ One assumption and one tradeoff
- ☐ Keep/change redesign for the new mission
- ☐ Evidence-based explanation
- ☐ One occupation and job connection